```
#include <iostream>              // Header file for input and output
using namespace std;             // Use standard namespace

int main()                       // Main function
{
   float discount;               // Variable to store discount percentage

   cout << "Enter discount percentage: "; // Ask user to enter discount
   cin >> discount;              // Accept discount from user

   try                           // Start try block
   {
      if (discount < 0 || discount > 100) // Check if discount is invalid
      {
         throw discount;         // Throw exception if outside 0-100
      }

      cout << "Valid Discount: " << discount << "%" << endl; // Display valid discount
   }

   catch (float d)               // Catch the thrown float exception
   {
      cout << "Exception: Invalid discount percentage!" << endl; // Display error
      cout << "Discount must be between 0 and 100." << endl; // Show valid range
   }

   return 0;                     // End the program
}
```